Appendix: Finding Overlapping Hyperbolic Block Communities

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A Abstract

This online appendix for the paper "XXX" contains supplementary information omitted from the main text due to space constraints.

B The Algorithm

Global likelihood defintion

Algorithm 1 Main Iterative Community Expansion

Require: Graph G, initial temperature T_0, temperature adjustment parameter α, maximum temperature $T_{\max}$, maximum number of iterations $R_{\max}$, batch size B, addition probability p_{add}

Compute initial communities C
$T \leftarrow T_0$
$r \leftarrow 0$
while $T < T_{\max}$ **and** $r < R_{\max}$ **do**
  $C_{\text{old}} \leftarrow C$
  $C \leftarrow \textsc{PrimaryExpansion}(G, C, T, B, p_{\text{add}})$
  if $\sum_{C \in \mathcal{C}} |C| \leq \sum_{C \in C_{\text{old}}} |C|$ **then**
    $T \leftarrow T + \alpha$
  else
    if $T > T_0$ **then**
      $T \leftarrow \max(T_0, T - \alpha)$
  $r \leftarrow r + 1$
return C

Algorithm 1 summarizes our iterative community expansion procedure. To this end, let C' denote a set of proposed community nodes obtained by either adding or removing a vertex from C. Following the likelihood formulation in (15) from [2], we evaluate a candidate update by considering only the likelihood terms affected by the modification of the current community. The corresponding objective is defined as

$$\ell(C' \mid C) = |E_{C'}| \log(d_{C'}) + |\overline{E_{C'}}| \log(1 - d_{C'}) + |E \setminus A_{C'}| \log(d'_O) + \left(|\overline{A_{C'}}| - |E \setminus A_{C'}|\right) \log(1 - d'_O), \quad (1)$$

where

$$A_{C'} = A_{C'} \cup \bigcup_{D \in C,\, D \neq C} A_D$$

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Algorithm 2 PrimaryExpansion

Require: Graph G, current communities C, temperature T, batch size B, addition probability p_{add}

for each community C in decreasing likelihood order **do**
  Construct candidate pool $\Gamma(C)$
  if $\Gamma(C) = \emptyset$ **then**
    continue
  $\mathcal{B} \leftarrow \emptyset$
  while $|\mathcal{B}| < B$ **do**
    Sample $u \sim \texttt{Uniform}(0, 1)$
    if $u \leq p_{\text{add}}$ and $\Gamma(C) \neq \emptyset$ **then**
      Sample $v \sim \texttt{Uniform}(\Gamma(C))$
      Add $(v, \textsc{Add})$ to $\mathcal{B}$
    else if $u > p_{\text{add}}$ and $C \neq \emptyset$ **then**
      Sample $v \sim \texttt{Uniform}(C)$
      Add $(v, \textsc{Remove})$ to $\mathcal{B}$
  Compute the partial log likelihood $\ell_{\text{cur}} = \ell(C \mid C)$ using (1)
  for each proposal $(v, q) \in \mathcal{B}$ **do**
    if $q = \textsc{Add}$ **then**
      $C' \leftarrow C \cup \{v\}$
    else
      $C' \leftarrow C \setminus \{v\}$
    Update (γ_C, H_C) and d_C for C'
    Reconstruct the community area $A_{C'}$ and d'_O
    Compute $\ell_{\text{new}} = \ell(C' \mid C)$
    $\Delta \leftarrow \ell_{\text{new}} - \ell_{\text{cur}}$
    if $\Delta \geq 0$ **then**
      $C \leftarrow C'$
      $\ell_{\text{cur}} \leftarrow \ell_{\text{new}}$
    else
      Sample $u_{\text{acc}} \sim \texttt{Uniform}(0, 1)$
      if $u_{\text{acc}} < \exp(\Delta / T)$ **then**
        $C \leftarrow C'$
        $\ell_{\text{cur}} \leftarrow \ell_{\text{new}}$
return C

denotes the union of the community areas after applying the proposed update and C is set of predicted community nodes. The global outside density is recomputed for each proposed state as

$$d'_O = \frac{|E \setminus A_{C'}|}{|\overline{A_{C'}}|}.$$

The quality of a proposed move is measured by the change in the objective,

$$\Delta = \ell(C' \mid C) - \ell(C \mid C).$$

A proposal that improves the likelihood, i.e., $\Delta \geq 0$, is always accepted. Otherwise, it is accepted according to the simulated-annealing probability $P_{\text{acc}} = \exp\left(\frac{\Delta}{T}\right), \Delta < 0$, where T denotes the temperature. This probabilistic acceptance of non-improving moves allows the search to escape poor local optima during community expansion. The procedure is summarized in Algorithm 2.

Table 1: Summary of the real-world networks.

Algor.	YT	Friend	DBLP	LJ	Amazon	Orkut
# nodes	19206	154300	16708	33568	3951	613943
# edges	241260	6159798	107626	2061290	25644	33343826
# com.	234	1567	47	502	48	4242
Alg. 1	234	1567	47	502	48	4242
Louvain	28	139	92	208	63	48
HyCoM-FIT (HyCom)	168	8047	48	60	15	23618

C Experimental Evaluation

C.0.1 Real-World Networks. We use six datasets from the Stanford Network Analysis Project [1]. We can describe the networks briefly as follows:

Youtube (YT) is a social network in which users establish friendship connections and can create groups that other users may join based on shared interests in video content. Friendster is an online gaming network that allows users form a group which other members can then join games. DBLP is a co-authorship network in computer science research papers. Two authors are connected if they publish at least one paper together. LiveJournal (LJ) and Orkut are social networking platforms where users establish friendship connections and create groups that other users can join. Amazon was collected by crawling Amazon website. In the resulting network, two products i and j are connected by an undirected edge if they are frequently purchased together.

From these networks, we retain only communities containing between 50 and 1000 nodes. The main characteristics of the resulting networks are summarized in Table 1.

C.1 Measures

Let $\mathcal{P}$ and $\mathcal{T}$ denote the collections of predicted and ground-truth community nodes, respectively. For a set of predicted community nodes $C_p \in \mathcal{P}$ and a set of ground-truth community nodes $C_t \in \mathcal{T}$, define $P(C_t, C_p) = \frac{|C_t \cap C_p|}{|C_p|}$ and $R(C_t, C_p) = \frac{|C_t \cap C_p|}{|C_t|}$.

The corresponding F_1 is $F_1(C_t, C_p) = \frac{2P(C_t,C_p)R(C_t,C_p)}{P(C_t,C_p)+R(C_t,C_p)}$. Then F_1^t Max are then defined as

$$F_1^t \text{ Max} = \frac{1}{|\mathcal{T}|} \sum_{C_t \in \mathcal{T}} \max_{C_p \in \mathcal{P}} F_1(C_t, C_p).$$

Let M denote the bipartite matching between the set of ground-truth communities $\mathcal{T}$ and the set of predicted communities $\mathcal{P}$, where the weight of edge (C_t, C_p) is $F_1(C_t, C_p)$.

Let $M_t^\star = \arg\max_M \sum_{(t,p)\in M} F_1(C_t, C_p)$ denote the maximum-weight one-to-one matching between ground-truth and predicted communities. Then F_1^t Mat score is defined as

$$F_1^t \text{ Mat} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \begin{cases} F_1(C_t, C_p), & \text{if } (t,p) \in M_t^\star, \\ 0, & \text{otherwise.} \end{cases}$$

Table 2: Parameter settings for synthetic and real-world experiments.

Parameter	Synthetic Net.	Real-world Net.
Representative Seed Selection		
Sampling ratio ρ_{rss}	0.15	0.3
Random-Walk Seed Selection		
Starting vertices	$4\lvert\mathcal{T}\rvert \ln \lvert\mathcal{T}\rvert$	$2\lvert\mathcal{T}\rvert \ln \lvert\mathcal{T}\rvert$
Walks per vertex	1000	1200
Maximum walk length L	50	150
Sampling ratio ρ_{rw}	0.10	0.30
Selected seeds	$\lvert\mathcal{T}\rvert$	$\lvert\mathcal{T}\rvert$

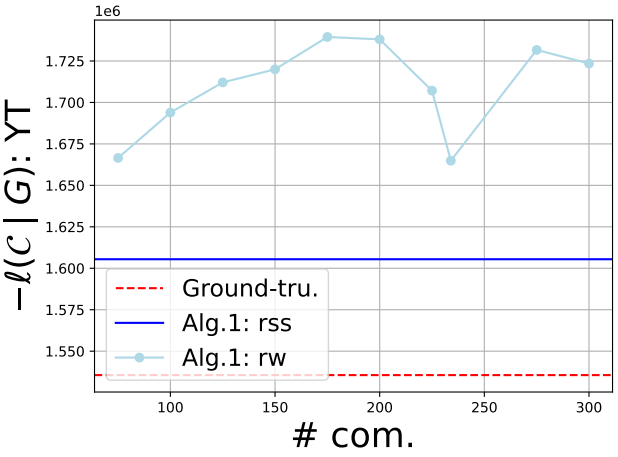


Figure 1: $-\ell(C \mid G)$ scores for Algorithm 1 with Random Walks (rw) by varying number of communities in YT data.

References

[1] Leskovec, J., Krevl, A.: SNAP Datasets: Stanford large network dataset collection. http://snap.stanford.edu/data (Jun 2014)

[2] Metzler, S., Günnemann, S., Miettinen, P.: Hyperbolae are no hyperbole: Modelling communities that are not cliques. In: 16th IEEE International Conference on Data Mining. pp. 330–339 (2016)